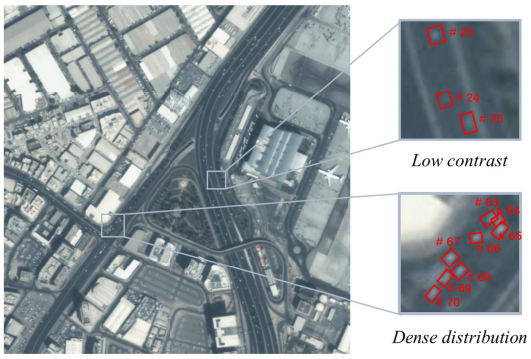
Oriented Vehicle Joint Detection and Tracking in Satellite Video via Identifier-Free Point Supervision

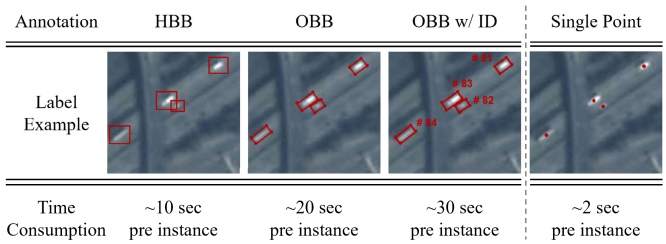
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***Abstract*—Oriented vehicle detection and tracking play a crucial role in various real-world applications. Yet, existing advanced models heavily rely on abundant and accurate oriented bounding box (OBB) and tracking identifier annotations, which are extremely labor-intensive for satellite videos. In this article, we endeavor to employ identifier-free point annotation to achieve competitive performance while minimizing annotation costs. Specifically, each instance across video frames is labeled by a single point, without providing its instance identifier. Building upon this setting, we introduce an oriented vehicle joint detection and tracking framework for satellite video, focusing on enhancing model performance by carefully designed sample acquisition and robust learning processes. First, we leverage temporal and visual information to generate sequence-aligned pseudo-labels and visually aligned synthetic objects, which complement each other during training by providing both exact appearance and annotation information. Second, a novel spatiotemporal consistency metric is developed to assess sample quality, which is then incorporated into a curriculum learning schedule. This strategy facilitates a gradual learning progression from high-quality data to low-quality or noisy examples, thereby minimizing interference from potentially misleading samples. Finally, an end-to-end oriented object joint detection and tracking network is constructed to enable effective oriented vehicle dynamic analysis. Extensive ablation and experimental results on two satellite video datasets demonstrate the superiority of our proposed method.**

***Index Terms*—Multiobject tracking (MOT), oriented vehicle detection, satellite video, weakly supervised learning.**



(a) Typical challenging scenarios in satellite video annotation



(b) Different annotations and the time costs

Fig. 1. Challenges and costs of satellite video annotation. (a) Typical difficult scenarios in satellite video annotation, including small-sized, blurry, and crowded objects. (b) Label examples in various annotation manners and their corresponding time costs. HBBs fail to tightly encapsulate vehicle objects, potentially causing overlap. OBB annotations are labor-intensive, and the task is further complicated by the requirement for tracking identifiers. In contrast, identifier-free annotation significantly reduces the annotation cost [10].

I. Introduction

CHARACTERIZED by wide coverage and large spatiotemporal information volume, satellite video has offered valuable dynamic insights into surface targets and has become an indispensable resource for geographic information systems. In the realm of satellite video analysis, oriented object detection [1], [2], [3] and tracking [4], [5], [6] stand out for their substantial applicability in real-world scenarios. For instance, in urban planning, it enables efficient analysis of traffic patterns and land-use dynamics to support infrastructure development; in border surveillance, it aids in detecting and predicting unauthorized border crossings and suspicious vehicular activities; and in disaster response, it facilitates rapid damage assessment and efficient coordination of rescue operations [7], [8], [9].

While deep learning has made remarkable strides in satellite video object detection and tracking, its efficacy is contingent upon an abundance of accurate data labels, which is extremely laborious. A single frame of satellite video can contain over a hundred objects due to the vast observation range, and the small objects often range in size from a few pixels to a dozen pixels and frequently appear blurred, especially against low-contrast backgrounds or when densely clustered [Fig. 1(a)]. This poses a substantial challenge for annotating oriented bounding boxes (OBBs), necessitating the careful adjustment of four points to align accurately with the object's edges. Additionally, the annotation of tracking identifiers (IDs) introduces further complexity, as researchers must continuously monitor and trace the trajectories of each object throughout the video sequence.

The contradiction between the growing demand for practical applications and the high cost of detailed annotation highlights

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the urgent need to develop low-cost alternatives. To minimize annotation effort as much as possible, we adopt identifier-free point annotations to construct a weakly supervised detection and tracking model. As shown in Fig. 1(b), this manner can reduce the time consumption by over 90%. Motivated by this advantage, the research community has begun to investigate point-supervised oriented object detection methods in high-resolution aerial imagery. For instance, methods such as Point2RBox [11] and PointOBB [12] leverage priors for data augmentation or pseudo-label generation. However, these priors are not well-suited for spotted vehicle objects in satellite videos. Other approaches like PBA [13], PMHO [14], and PointSAM [15] employ SAM [16] to generate target masks and then derive pseudo-OBBs. Nevertheless, SAM's segmentation performance is significantly limited when processing small and blurry objects, thereby rendering these methods incapable of generating accurate pseudo-labels in satellite videos.

In the field of multiobject tracking (MOT), state-of-the-art methods such as FairMOT [17], CenterTrack [18], and TraDeS [19] achieve high performance but rely heavily on extensive and costly annotations. On the other hand, many weakly supervised or unsupervised approaches, including SORT [20] and ByteTrack [21], adhere to the tracking-by-detection paradigm. These methods rely on bounding boxes predicted by a detector and associate them using motion models. Nevertheless, this paradigm suffers from inefficient feature utilization due to the separation of detection and tracking processes. This inefficiency is particularly pronounced in satellite videos, which often contain dense and occluded objects, and any deficiencies in the detector can significantly impair tracking performance.

Regarding the above issues, we proposed an identifier-free point supervised oriented vehicle joint detection and tracking framework, termed P-OVDT, in this article. To elaborate, P-OVDT explores temporal and apparent information to generate refined pseudo-labels and synthetic objects. It starts by employing unsupervised methods for foreground extraction and association, followed by the estimation of pseudo-OBBs by aligning parameters along the sequence dimension. Additionally, synthetic objects are generated based on real vehicle visual patterns using a Gaussian sampling strategy and placed onto clean backgrounds. These synthetic objects, with precise labels, compensate for the inaccuracies in pseudo-labels, while the inclusion of real objects mitigates overfitting to the synthetic objects. For the training methodology, P-OVDT employs a curriculum learning schedule based on a novel spatiotemporal consistency metric. This metric assigns higher confidence to objects exhibiting consistent OBBs over long-term tracking sequences. Consequently, the training phase begins with high-quality samples and gradually incorporates lower-quality ones, thereby minimizing the interference of noisy samples during feature extraction. Architecturally, considering the scarce appearance information of satellite video vehicles, which makes it difficult to extract discriminative ReID features, the embedding branch encodes the objects' spatial distribution pattern as a vital supplement by a customized spatial pooling module (SPM).

In summary, the main contributions of this article are as follows.

1) To the best of our knowledge, this article represents a pioneering study on oriented vehicle detection and tracking in satellite video based on identifier-free point annotations.
2) The temporal and visual patterns are thoroughly exploited to produce high-quality pseudo-labels and synthetic objects, which collectively serve as complementary guidance for the weakly supervised learning process.
3) A customized spatiotemporal consistency metric is designed to evaluate sample quality, enabling the network to gradually incorporate more complex spatial–temporal patterns within the curriculum learning schedule.
4) Comprehensive experimentation confirms that the proposed method significantly outperforms existing point-supervision methods. It demonstrates a performance degradation of less than 20% when compared to fully supervised methods, while incurring only 6% of the annotation costs.

II. Related Work

A. Oriented Object Detection in Aerial Imagery

Oriented object detection significantly advances aerial image analysis by tightly outlining objects with diverse orientations and high aspect ratios. For an image I, the task is to predict a set of OBBs $\mathbf{B} = \{b_i\}$, where each $b_i = (x_i, y_i, w_i, h_i, \theta_i)$ denotes the center coordinates, width, height, and orientation angle of an object. State-of-the-art methods have achieved promising advances through several aspects: developing novel network architectures (e.g., S^2A-Net [22], R^3Det [23], and SCRDet++ [24] among CNN-based models; AO^2-DETR [25], Oriented-DETR [26], and ARS-DETR [27] from DETR-based variants), designing adaptive object representations (such as Oriented RepPoints [28], Gliding Vertex [29], and MidNet [3]), and introducing specialized loss functions (including GWD [30] and KFIoU [31]).

With the progress achieved by the aforementioned methods, the high cost of annotating OBBs has become a major constraint limiting their broader practical application. As a result, researchers have begun to explore weakly supervised methods. H2RBox [32], H2RBox-v2 [33], EIE-Det [34], and AFWS [35] utilized horizontal bounding boxes (HBBs) for supervision, generating oriented boxes. Subsequently, methods like PointOBB [12], Point2RBox [11], and P2RBox [10] employed point annotations for OBB generation. These methods typically employ a sequence of augmentations, including flips, rotations, and scaling, on the objects, and engage in self-supervised learning based on the consistency of the data before and after these transformations. However, the small-scale and blurry objects commonly found in satellite videos are less responsive to these transformations compared to larger objects, which can impede the extraction of robust features. Furthermore, these methods are primarily designed for static

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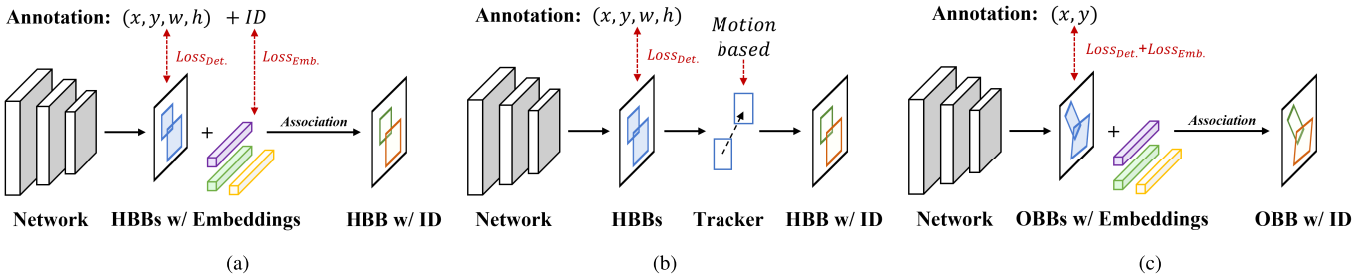


Fig. 2. Classical paradigms for various object detection and tracking methods. (a) Mainstream joint detection and embedding methods exhibit high feature utilization efficiency and performance but necessitate extensive annotation. Some motion-based unsupervised (b) tracking-by-detection methods can eliminate the need for ID annotations. Yet, their performance is heavily contingent on the accuracy of the detector. (c) Our method is dedicated to utilizing only position information to learn OBBs and ReID features in an end-to-end framework, enhancing model efficiency while minimizing annotation costs to the greatest extent.

images and overlook the temporal information within video sequences.

Furthermore, SAM's [16] robust zero-shot segmentation capability offers a promising alternative for weakly supervised object detection. Methods such as PBA [13], PMHO [14], and PointSAM [15] incorporate SAM to support OBB learning from point annotations. They utilize SAM to produce object masks from images, thereby enhancing weakly supervised feature learning. However, in very low-resolution satellite videos, the masks generated by SAM are often ineffective, and these methods remain highly susceptible to noise.

B. Multiple Objects Tracking in Satellite Video

MOT addresses the challenging problem of locating multiple targets and maintaining their identities across frames. Given a satellite video sequence $\mathbf{V} = \{I^1, I^2, \ldots, I^T\}$ of T frames, the goal is to estimate a set of bounding boxes $\mathbf{B}^t = \{b_i^t\}$ and their identities $\mathrm{ID_i}$ for each frame I^t. In natural videos, many advanced MOT methods such as FairMOT [17], JDE [36], CenterTrack [18], and TrackFormer [37] have achieved impressive performance by leveraging strong appearance features and complex data association strategies. However, these approaches predominantly rely on rich visual appearance cues to extract discriminative reidentification features, which exhibit limited generalization in satellite videos where objects are small, blurry, and lack sufficient appearance details.

Trackers specifically tailored for satellite video have been introduced to address these challenges. The CFTracker [4] introduces a novel cross-frame training process, jointly considering feature outputs across multiple frames to capture motion features of specific objects. SMTNet [5] encodes the coordinate information of nearby objects to build a spatial LSTM for extracting positional information. GAMO [6] constructs a graph model where nodes represent target objects and edges denote spatial relationships between them, thereby capturing spatial positional relationships between objects. While these deep learning methods have demonstrated performance enhancements in satellite video analysis, the annotation costs required are extremely high. More importantly, these methods are tailored for HBBs and are difficult to predict the OBB trajectories, thus limiting their ability to accurately represent the dynamic information.

In contrast, unsupervised tracking methods, such as KCF [38] and SORT [20], typically follow the tracking-by-detection paradigm, relying on predicted detection boxes for object association. These approaches are limited by their weak feature representation capacity, and their performance degrades severely when the detection results are unreliable. The dense and crowded scenes commonly found in satellite videos pose significant challenges to the effectiveness of such methods.

III. Preliminary and Problem Definition

A. Preliminary on Joint Object Detection and Tracking

Joint multiobject detection and tracking targets for localizing a set of objects and assigning a unique identifier for each instance simultaneously. Without loss of generality, a prevailing approach for joint multiobject detection and tracking is to integrate a separate ReID head, for instance, embedding extraction on top of a detection network. This enables the temporal propagation of instance identifiers by comparing interframe embedding similarities. As illustrated in Fig. 2(a), a video frame I^t is first processed through a feature extractor to generate the feature map $\mathcal{F}^t$. Then, $\mathcal{F}^t$ is fed into the prediction head Ψ to simultaneously produce detections $\hat{\mathbf{B}}_H^t$ and their associated instance embeddings $\hat{\mathbf{E}}^{\mathbf{t}}$

$$\left[\hat{\mathbf{B}}_H^t, \hat{\mathbf{E}}^t\right] = \Psi\left(\mathcal{F}^t\right) \tag{1}$$

in which $\hat{\mathbf{B}}_H^t$ denotes the set of detected horizontal boundary boxes at time stamp t, $\hat{\mathbf{B}}_H^t = \{\hat{b}_{H,1}^t, \hat{b}_{H,2}^t, \ldots\}$. Each element $\hat{e}_i^t$ in $\hat{\mathbf{E}}^{\mathbf{t}}$ is associated with a detected object in $\hat{\mathbf{B}}_H^t$, and it corresponds to a C-dimensional vector. Finally, the boxes $\hat{\mathbf{B}}_H^t$ and the embeddings $\hat{\mathbf{E}}^{\mathbf{t}}$ are associated with existing trajectories using bipartite matching Θ

$$\left[\hat{\mathbf{T}}^{\mathbf{t}}\right] = \Theta\left[\left(\hat{\mathbf{B}}_H^t, \hat{\mathbf{E}}^{\mathbf{t}}\right), \hat{\mathbf{T}}^{\mathbf{t-1}}\right] \tag{2}$$

where $\hat{\mathbf{T}}^{\mathbf{t}} = \{\hat{T}_1^t, \hat{T}_2^t, \ldots\}$ and $\hat{T}_{\mathrm{ID}}^t = \{\hat{b}_{H,\mathrm{ID}}^1, \ldots, \hat{b}_{H,\mathrm{ID}}^t\}$ denotes the trajectory with the identifier as ID, consisting of the associated bounding boxes from previous frames up to the tth frame.

To obtain a reliable multiobject tracker, a combination of loss functions is used to guide the optimization of model parameters as

$$\begin{aligned}\mathcal{L} = {} & \mathcal{L}_{\mathrm{loc}}\left[(\hat{x}, \hat{y}), (x, y)\right] \\ & + \mathcal{L}_{\mathrm{size}}\left[\left(\hat{w}, \hat{h}\right), (w, h)\right] + \mathcal{L}_{\mathrm{emb}}\left(\hat{e}, \mathrm{ID}\right)\end{aligned} \tag{3}$$

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where $\mathcal{L}_{\text{loc}}$ and $\mathcal{L}_{\text{size}}$ are responsible for improving the localization accuracy and $\mathcal{L}_{\text{emb}}$ aims to enhance the similarity between embeddings belonging to the same instance.

B. Problem Definition of Joint Object Detection and Tracking Under Identifier-Free Point Annotation

The above learning paradigm is based on the assumption that the OBB annotations of objects and their associated instance identifiers are readily available during the training stage, which is highly labor-intensive. Therefore, joint oriented object detection and tracking under identifier-free point supervision is defined in this section.

First, our task expects *Head* network Ψ to additionally learn the orientation features of the objects and generate OBBs, that is,

$$\left[\hat{\mathbf{B}}_O^t, \hat{\mathbf{E}}^t\right] = \Psi\left(\mathcal{F}^t\right) \tag{4}$$

where $\hat{\mathbf{B}}_O^t = \{\hat{b}_{O,1}^t, \hat{b}_{O,2}^t, \ldots, \}$, with each OBB $\hat{b}_{O,i}^t = \{\hat{x}_i^t, \hat{y}_i^t, \hat{w}_i^t, \hat{h}_i^t, \hat{\theta}_i^t\}$ containing not only position and size information but also additional angle $\hat{\theta}_i^t$ to enable a more accurate description of the object. Data association is then performed using $\hat{\mathbf{B}}_O^t$ and their corresponding embeddings $\hat{E}^t$

$$\left[\hat{\mathbf{T}}^{\mathbf{t}}\right] = \Theta\left[\left(\hat{\mathbf{B}}_O^t, \hat{\mathbf{E}}^{\mathbf{t}}\right), \hat{\mathbf{T}}^{\mathbf{t-1}}\right]. \tag{5}$$

Meanwhile, to minimize annotation costs as much as possible, we just labeled a single point within the object, providing only (x, y) as supervision, without offering ground truth for size (w, h), angle θ, and identifier id. Consequently, our loss function is defined as

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{loc.}}\left[(\hat{x}, \hat{y}), (x, y)\right] + \mathcal{L}_{\text{size.}}\left[\left(\hat{w}, \hat{h}\right), \left(\tilde{w}, \tilde{h}\right)\right] \\ + \mathcal{L}_{\text{ang.}}\left(\hat{\theta}, \tilde{\theta}\right) + \mathcal{L}_{\text{emb.}}\left(\hat{e}, \tilde{\text{ID}}\right)\end{aligned} \tag{6}$$

where $\tilde{w}$, $\tilde{h}$, $\tilde{\theta}$, and $\tilde{\text{ID}}$ represent the pseudo-labels that are self-generated by the model.

To sum up, we aim to achieve efficient dynamic analysis of satellite video objects with minimal annotation costs. Based on our problem definition, the three-level challenges compared to previous work are as follows.

1) Using only point annotations to learn the OBB information of blurry small objects.
2) Learning ReID embeddings in an identifier-free way, that is, without using any temporal annotation information.
3) Attempting to track oriented objects instead of horizontal ones and jointly learn the parameters of OBB and ReID embeddings.

To this end, we advocate obtaining high precision under identifier-free point supervision, as detailed in the following.

IV. Proposed Method

In this section, we propose P-OVDT, a novel framework for jointly training oriented object detection and MOT in satellite video. The overall pipeline is illustrated in Fig. 3. Given consecutive video frames as input, the network first extracts spatiotemporal features through a shared backbone. The detection head then branches into four parallel subnetworks: position, size, orientation, and embedding. Beyond leveraging identifier-free point annotations for weak supervision, P-OVDT further generates high-quality pseudo-labels and synthetic objects with precise ground-truth labels. During training, we introduce a spatiotemporal consistency measure, which is integrated into a curriculum learning strategy to progressively refine model predictions.

A. Sequence-Aligned Pseudo-Label Generation

In this section, we aim to generate high-quality OBB trajectories from identifier-free point annotations. Some weakly supervised video models initially generate pseudo-bounding boxes by distinguishing the moving foreground from the background through background modeling. However, this method is not suitable for our task, as illustrated in Fig. 4, blobs can be inaccurate when objects are dim and small, and a tiny number of mistakenly extracted foreground pixels can lead to significant degradation in orientation estimation accuracy. To address this issue, we design a precise OBB-trajectory estimation method by aligning pseudo labels across video sequences in this section, and the pipeline is shown in Fig. 5.

Initially, background subtraction method is applied to video frame I^t to distinguish moving foregrounds $\mathbf{F}^t = \{f_1^t, f_2^t, \ldots\}$ from stationary backgrounds. And then, if a labeled point (x_i^t, y_i^t) lies within a contiguous region f_i^t, f_i^t is retained, and its circumscribed HBB $\tilde{b}_{H,i}^t = \{x_i^t, y_i^t, \tilde{w}_{H,i}^t, \tilde{h}_{H,i}^t\}$ is extracted as the pseudo-HBB of the ith object in the tth frame. Otherwise, the foreground f_i^t is discarded. After traversing all annotation points, these pseudo-HBBs $\tilde{\mathbf{B}}_H^t$ are then input into the motion-based tracker Θ to get the pseudo-HBB tracklets $\tilde{\mathbf{T}}_H^t$

$$\left[\tilde{\mathbf{T}}_H^t\right] = \Theta\left[\tilde{\mathbf{B}}_H^t, \tilde{\mathbf{T}}_H^{t-1}\right]. \tag{7}$$

Next, we derive high-quality pseudo-OBB trajectories from pseudo-HBBs by aligning object posture across the sequence. The tracking results, fortunately, offer a pronounced prior, a benefit derived from the bird's-eye view perspective, which ensures a close alignment between vehicle movement and their orientation. Therefore, by calculating the displacement of an object's center point within tracklets labeled ID between adjacent frames, the rotation angle $\tilde{\theta}_{\text{id}}^t$ can be readily determined. Specifically

$$\tilde{\theta}_{\text{id}}^{t-1,t} = \cos^{-1}\left(\frac{x_{\text{id}}^t - x_{\text{id}}^{t-1}}{\sqrt{\left(x_{\text{id}}^t - x_{\text{id}}^{t-1}\right)^2 + \left(y_{\text{id}}^t - y_{\text{id}}^{t-1}\right)^2}}\right) \tag{8}$$

$\tilde{\theta}_{\text{id}}^t$ is the average of $\tilde{\theta}_{\text{id}}^{t-1,t}$ and $\tilde{\theta}_{\text{id}}^{t,t+1}$ to smooth the angles in the adjacent three frames.

In terms of size, based on the rotation angle $\tilde{\theta}_{\text{id}}^t$ and its pseudo-HBB $\tilde{b}_{H,\text{id}}^t$, then $\tilde{w}_{O,\text{id}}^t$ and $\tilde{h}_{O,\text{id}}^t$ for the pseudo-OBB can be derived

$$\begin{aligned}\tilde{w}_{H,\text{id}}^t &= \tilde{h}_{O,\text{id}}^t \cdot \sin\left(\tilde{\theta}_{\text{id}}^t\right) + \tilde{w}_{O,\text{id}}^t \cdot \cos\left(\tilde{\theta}_{\text{id}}^t\right) \\ \tilde{h}_{H,\text{id}}^t &= \tilde{h}_{O,\text{id}}^t \cdot \cos\left(\tilde{\theta}_{\text{id}}^t\right) + \tilde{w}_{O,\text{id}}^t \cdot \sin\left(\tilde{\theta}_{\text{id}}^t\right).\end{aligned} \tag{9}$$

However, as shown in Fig. 6(a), these equations alone do not yield a unique solution for $\tilde{w}_{O,\text{id}}^t$ and $\tilde{h}_{O,\text{id}}^t$. When necessary, additional prior constraints on vehicle targets can be leveraged

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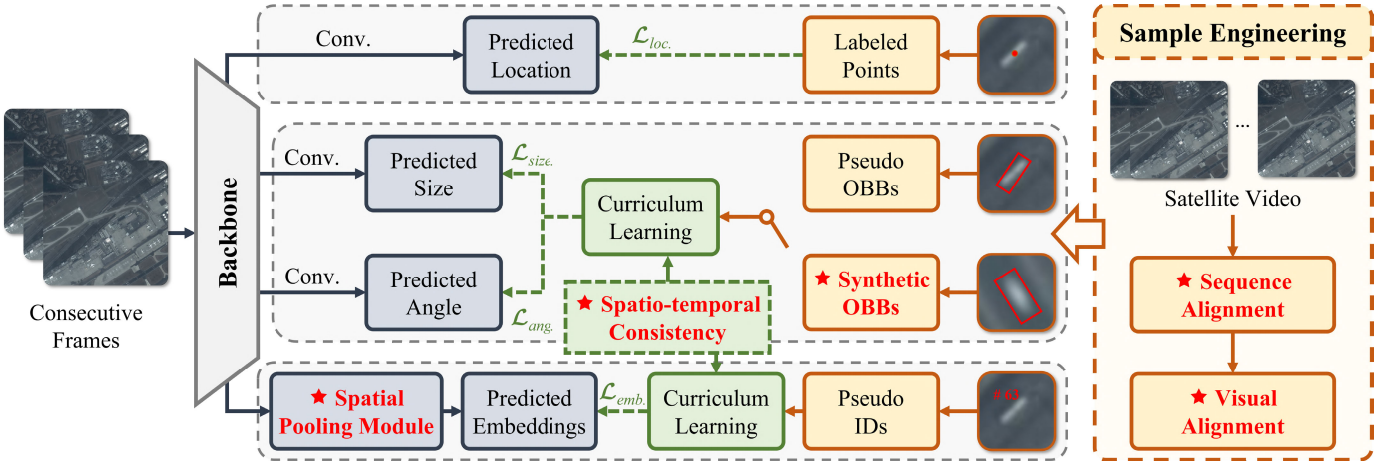


Fig. 3. Overview of the proposed P-OVDT. Consecutive satellite video frames are processed by a shared backbone, predicting location, size, angle, and embeddings. A novel SPM captures spatial patterns in embeddings. Sample engineering generates sequence-aligned pseudo-labels and visual-aligned synthetic objects. Curriculum learning, guided by spatiotemporal consistency, is utilized to bolster the stability of the convergence.

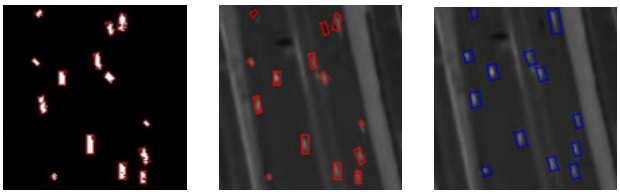


Fig. 4. Results of pseudo-OBBs estimation. (a) Foreground and background segmentation and the minimum enclosing boxes of foreground blobs. (b) Overlays these bounding boxes onto the original frame, leading to substantial errors with small, blurry objects in satellite videos. (c) Pseudo-OBBs generated by our proposed approach.

in satellite videos. First, we assume that small vehicles typically exhibit a specific aspect ratio k (commonly around 2.5), which is

$$\tilde{h}^t_{O,\mathrm{id}} = k \cdot \tilde{w}^t_{O,\mathrm{id}}. \tag{10}$$

Then, by combining (9) and (10), we can solve for

$$\begin{aligned} \tilde{w}^t_{O,\mathrm{id}} &= \frac{\tilde{w}^t_{H,\mathrm{id}}}{k \cdot \sin\left(\tilde{\theta}^t_{\mathrm{id}}\right) + \cos\left(\tilde{\theta}^t_{\mathrm{id}}\right)} \\ \tilde{h}^t_{O,\mathrm{id}} &= \frac{k \cdot \tilde{w}^t_{H,\mathrm{id}}}{k \cdot \sin\left(\tilde{\theta}^t_{\mathrm{id}}\right) + \cos\left(\tilde{\theta}^t_{\mathrm{id}}\right)}. \end{aligned} \tag{11}$$

Second, we recognize that some larger trucks may have an aspect ratio significantly greater than k; however, since the width of the road is finite, the width of these objects usually does not exceed a certain upper limit $W_{\max}$ (usually 2.5 m). Therefore, we introduce an upper bound $\tilde{w}_{\max}$, which is the ratio of $W_{\max}$ to the ground sample distance (GSD) of the video. If the $\tilde{w}^t_{O,\mathrm{id}}$ obtained from (11) exceeds $\tilde{w}_{\max}$, the result will be adjusted accordingly

$$\begin{aligned} \tilde{w}^t_{O,\mathrm{id}} &= \tilde{w}_{\max} \\ \tilde{h}^t_{O,\mathrm{id}} &= \frac{\tilde{w}^t_{H,\mathrm{id}} - \tilde{w}_{\max} \cdot \cos\left(\tilde{\theta}^t_{\mathrm{id}}\right)}{\sin\left(\tilde{\theta}^t_{\mathrm{id}}\right)}. \end{aligned} \tag{12}$$

Thus, as illustrated in Fig. 6, the size of the object can be estimated, leading to the acquisition of complete pseudo-OBB $\tilde{b}^t_{O,\mathrm{id}} = \{x^t_{\mathrm{id}}, y^t_{\mathrm{id}}, \tilde{w}^t_{O,\mathrm{id}}, \tilde{h}^t_{O,\mathrm{id}}, \tilde{\theta}^t_{\mathrm{id}}\}$. Finally, updating $\tilde{\mathbf{B}}^t_H$ in (7) with $\tilde{\mathbf{B}}^t_O$ yields the pseudo-OBB trajectories $\tilde{\mathbf{T}}^t_O$.

However, it is important to note that there are still some exceptional cases that need to be considered specifically. First, if an annotation point (x^t_i, y^t_i) does not match with the foregrounds $\mathbf{F}^t$, the size $(\tilde{w}^t_{O,\mathrm{id}}, \tilde{h}^t_{O,\mathrm{id}})$ of pseudo-OBB at that point is assigned the average value in the tth frame, and the angle $\tilde{\theta}^t_{\mathrm{id}}$ is set to that of the nearest object. This assumption is grounded in the probability that closely located vehicles are on the same road and thus share similar orientations. Moreover, if an object fails to associate sequentially, its angle is also estimated in the same manner.

B. Visual-Aligned Synthetic Object Generation

Note that, despite the full utilization of temporal information to generate fine pseudo-labels, failed associations can still lead to inaccurate pseudo-label estimation. Additionally, outliers may not adhere to the constraints described in (11) and (12). To address these issues, P-OVDT employs a method for synthesizing objects that closely mimic the visual patterns of real vehicles, thereby producing highly realistic samples with precise OBB information.

Given that vehicle objects in satellite videos appear as blurred and spotted, we employ ellipses with an internal grayscale governed by a Gaussian distribution to simulate them. Specifically, for an object template with an estimated width $\tilde{w}_0$ and a height $\tilde{h}_0$ (corresponding to the major and minor axes of the ellipse), the internal gray value is computed as follows:

$$\begin{aligned} g_{(x,y)} &= \tilde{g}_0 \cdot \left[1 - \exp\left(-\frac{\left(1 - d_{(x,y)}\right)^2}{2\sigma_e^2}\right)\right] \\ d_{(x,y)} &= \sqrt{\frac{(x - x_0)^2}{\tilde{w}_0} + \frac{(y - y_0)^2}{\tilde{h}_0}} \end{aligned} \tag{13}$$

where $g_{(x,y)}$ signifies the gray value at the coordinates (x, y), $\tilde{g}_0$ refers to the gray value at the template's center, σ_e denotes the standard deviation of the Gaussian function, which governs the rate of decay in gray values, and d represents the distance

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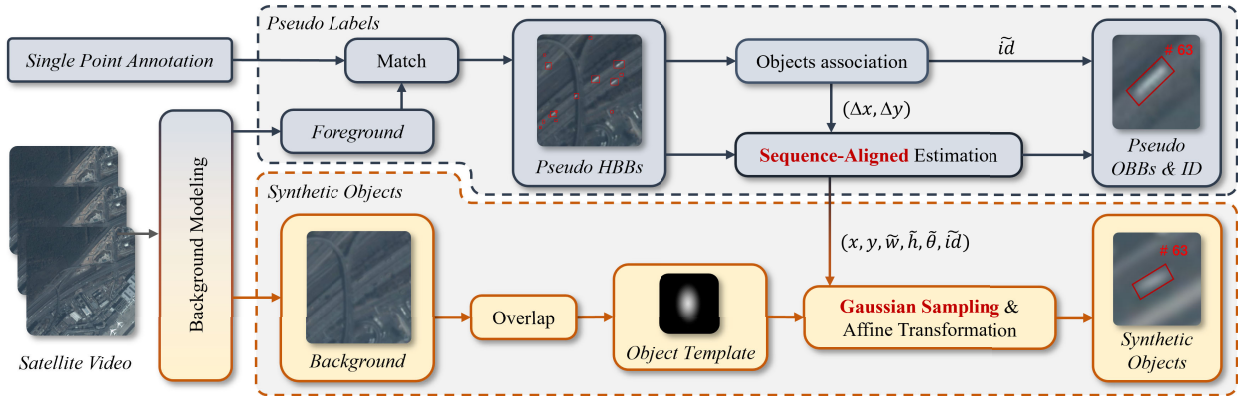


Fig. 5. Sample engineering in P-OVDT. The dark blue branch symbolizes the pseudo-label generation process, whereas the orange branch denotes the synthetic object generation process. These two branches collaborate to provide training samples that jointly participate in the weakly supervised learning process.

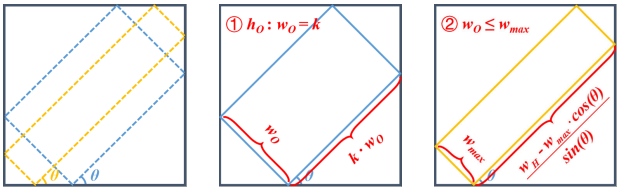


Fig. 6. Solving for the size of the pseudo-OBB under different constraint conditions. In (a), it can be observed that based on a determined circumscribed rectangle and rotation angle, a unique size for the OBB cannot be estimated. (b) Calculation method for small vehicles, where the aspect ratio of the OBB adheres to k. However, some large trucks may possess a larger aspect ratio; consequently, for these vehicles, as depicted in (c), we establish an upper limit for the width and subsequently calculate the height based on the maximum width.

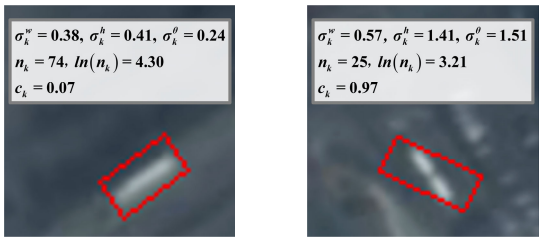


Fig. 7. Difficulty measures of different objects. It can be seen that, based on the designed spatiotemporal consistency measure, samples with accurate pseudo-labels exhibit lower difficulty scores and can be learned earlier. (a) Accurate pseudo-labels with lower difficulty measures. (b) Noisy sample with higher difficulty measures.

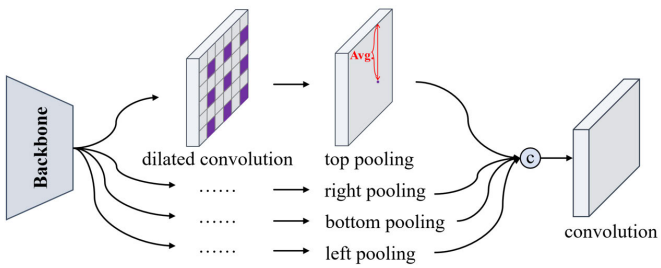


Fig. 8. Structure of the SPM. Four directional pooling operations are implemented to capture the spatial distribution patterns between objects.

from point (x, y) to the ellipse's boundary. An example for the object template is provided in Fig. 5.

Subsequently, the object templates are tiled at the positions of each annotation point (x_i^t, y_i^t), and they are ultimately transformed into synthetic objects through affine transformations. To enhance the diversity, the parameters of these synthetic objects adhere to Gaussian distributions, which can be formulated as

$$\left[\tilde{w}_0 \ \ \tilde{h}_0 \ \ \tilde{\theta}_0 \ \ \tilde{g}_0\right] \sim \mathcal{N}_4\left(\begin{bmatrix}\tilde{w}_{O,i}^t \\ \tilde{h}_{O,i}^t \\ \tilde{\theta}_{O,i}^t \\ \tilde{g}_{O,i}^t\end{bmatrix}, \begin{bmatrix}\sigma_s^2 & & & \\ & \sigma_s^2 & & \\ & & \sigma_s^2 & \\ & & & \sigma_s^2\end{bmatrix}\right) \tag{14}$$

where $\tilde{g}_i^t$ corresponds to the grayscale at (x_i^t, y_i^t) in the t-frame. σ_s^2 represents the variance of the Gaussian distribution, controlling the degree of randomness in the synthetic objects. Additionally, the tracking IDs of the synthetic objects are aligned with the pseudo-OBB trajectories.

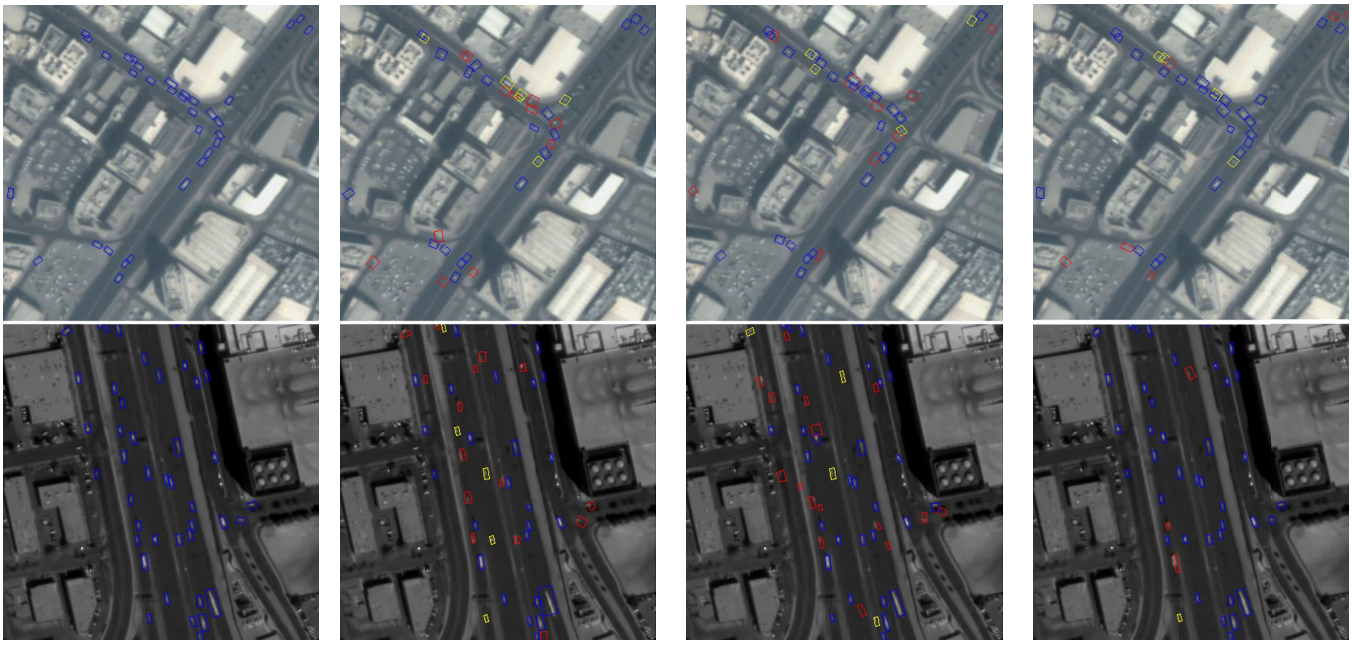
Therefore, this method produces visually aligned synthetic samples that act as supplementary learning cues, thereby mitigating the disruptive effects of inaccurate pseudo-labels. Concurrently, the integration of real samples prevents potential overfitting that could occur if synthetic objects were used exclusively. Throughout the training process, both types of samples are randomly selected to participate.

C. Spatiotemporal Consistency-Based Curriculum Learning

While P-OVDT has established methods for generating precise pseudo-labels and synthetic objects, it is acknowledged that in intricate scenarios such as heavy occlusion or dense crowds, there may still be inaccurate or missing OBBs and trajectories. Consequently, we devise a curriculum learning strategy grounded in spatiotemporal coherence to enhance the robustness of weakly supervised learning. Drawing on the principle of learning from easy to hard examples, curriculum learning [39], [40] sequentially organizes training samples by increasing difficulty to facilitate model convergence and improve generalization. This paradigm has been widely adopted in computer vision, particularly in weakly supervised settings [41], [42], [43].

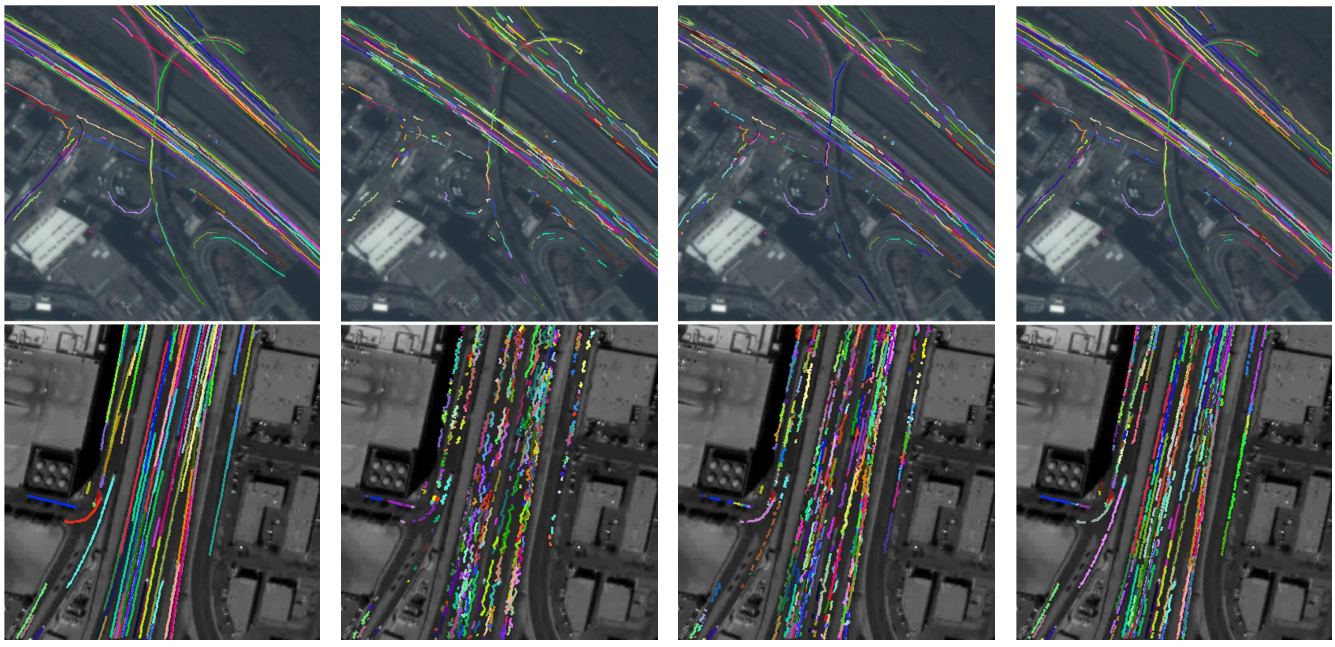
The core challenge in curriculum learning is designing an effective sample selection strategy. In our task, to screen high-quality pseudo-labels, the width, height, and orientation

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(a) (b) (c) (d)

Fig. 9. Ground-truth and visualization results of point-supervised detectors. The scenes in the first row are from the Jilin-1 dataset, while the last row showcases scenes from the Skysat dataset. The true positive is represented in blue, the false negative is in yellow, and the false positive is in red. (a) Ground truth. (b) PointOBB. (c) Point2RBox. (d) P-OVDT (Ours).



(a) (b) (c) (d)

Fig. 10. Trajectories generated by different detectors and trackers, with (a) being the ground truth. The completeness of a trajectory directly correlates with the accuracy of data association, and a smoother trajectory reflects the precision of the objects' location. (a) Ground truth. (b) PointOBB. (c) Point2RBox. (d) P-OVDT (Ours).

of vehicles must exhibit minimal variance throughout their trajectory. Moreover, longer trajectories, characterized by robust association, provide higher confidence in this consistency. Based on this consideration, we constructed the difficulty measures based on the spatiotemporal consistency as follows:

$$c_i = \varphi\left[\frac{\sigma_i^w + \sigma_i^h + \sigma_i^\theta}{3 \cdot \ln(n_i)}\right] \tag{15}$$

in which c_i serves as the difficulty measure for the ith sample, n_i represents the number within its corresponding trajectory, while σ_i^w, σ_i^h, and σ_i^θ are, respectively, the standard deviations of the width, height, and angles of the n_i samples across its trajectory, as longer trajectories inherently exhibit stronger association confidence. Additionally, φ is the linear normalization function.

Consequently, c_i is considered in the curriculum learning process, which is controlled by a function of the epoch

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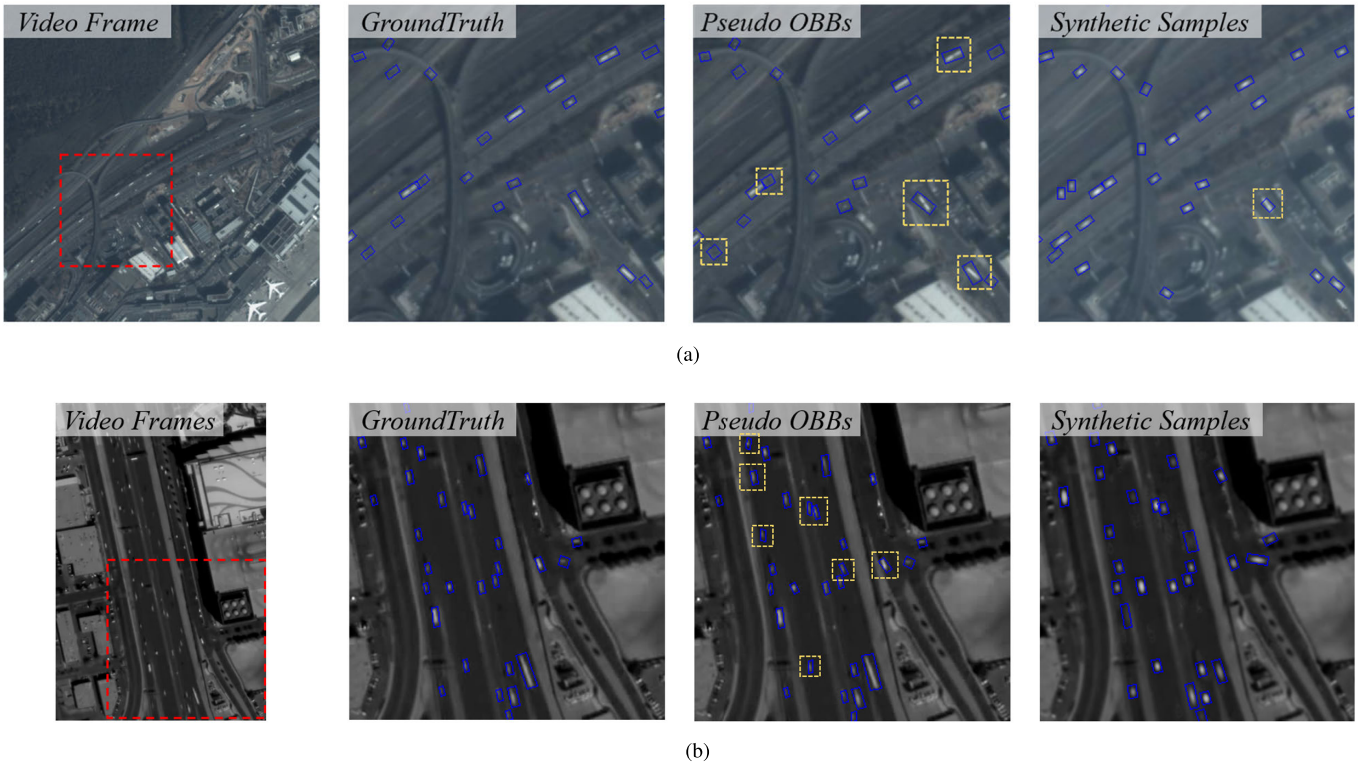


Fig. 11. Visualization of data generation. Noise samples are marked with yellow dashed boxes. (a) Video-009. (b) Video-012.

number p. In this article, we employ the geometric progression function [44]

$$\lambda(p) = \min\left(1, 2^{\left(\frac{\log_2 1 - \log_2 \lambda_0}{p_{\max}} p + \log_2 \lambda_0\right)}\right) \tag{16}$$

where λ_0 represents the initial difficulty threshold and $p_{\max}$ denotes the epoch when the function reaches 1 for the first time. During the pth epoch, samples with c_i below $\lambda(p)$ are selected for training, thereby regulating the curriculum learning process. Compared to other pacing functions, this schedule ensures that the network initiates training with straightforward yet high-quality examples for a longer duration, thereby guaranteeing the acquisition of more robust features during the early training phase.

Fig. 7 displays the difficulty measures for different samples. It is evident that by evaluating the spatiotemporal consistency, samples with precise pseudo-labels are assigned a lower learning difficulty. It is important to note that for all synthetic objects, which have meticulous labels, their difficulty measures are fixed at λ_0, facilitating the provision of sample pattern insights during the early epochs of training.

D. Model Implementation

Ultimately, leveraging the aforementioned sample generation methods and curriculum learning strategies, we constructed an end-to-end network for weakly supervised oriented vehicle joint detection and tracking. Given the small size of vehicle objects and the point annotations, the network architecture is based on the classical keypoint-based detector, CenterNet [45]. As depicted in Fig. 3, our network extends beyond the conventional location and size branches, incorporating angle and embedding branches. Following established practices in [5], [46], and [47] for capturing and preserving feature integrity for small objects, P-OVDT's backbone network adopts two architectural adaptations: 1) its first convolutional layer accepts three-frame stacked inputs (equivalent to nine channels for optical satellite videos) and 2) subsequent layers employ a conservative 2× downsampling rate for feature maps as opposed to the conventional 4× downsampling.

1) Spatial Pooling Module: For the embedding branch, the small size and blurriness of objects in satellite videos pose significant challenges for extracting discriminative appearance features. Fortunately, the bird's-eye-view perspective provides a distinct advantage by clearly revealing the spatial distribution of objects, offering a vital cue for association.

To leverage this, we design a custom SPM within the embedding branch, as detailed in Fig. 8. The SPM takes the feature map from the backbone and processes it through four parallel branches. Each branch employs a 3×3 dilated convolution (dilation rate as 2) to expand the receptive field, followed by corner pooling [48] in the left, top, right, and bottom directions. Specifically, the corner pooling operation traverses pixels along a specified direction (e.g., rightward along each row) and computes the average response within that direction, capturing contextual information about surrounding objects. The resulting features from the four directions are concatenated and then integrated through a convolutional block consisting of a 3×3 layer and a 1×1 layer for dimensionality reduction and feature aggregation. Through this design, the model significantly enlarges its receptive field and gains global awareness of object distribution patterns, enabling the

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network to learn more discriminative feature representations for embedding.

2) Training Loss: The loss of the P-OVDT is composed of the following four parts:

$$\mathcal{L} = \mathcal{L}_{\text{loc.}} + \alpha\mathcal{L}_{\text{size.}} + \beta\mathcal{L}_{\text{ang.}} + \gamma\mathcal{L}_{\text{emb.}} \tag{17}$$

where α, β, and γ are weight coefficients.

Consistent with CenterNet, the localization branch yields a heatmap that encapsulates center point information. The localization loss function $L_{\text{loc.}}$ also adopts the improved focal loss. For the size and angle branches, P-OVDT utilizes a weighted $L1$ loss to supervise the regression

$$\begin{aligned}\mathcal{L}_{\text{size.}} &= \frac{1}{N^p}\sum_{i=1}^{N^p}(1-c_i)\left|\left(\hat{w}_{O,i},\hat{h}_{O,i}\right)-\left(\tilde{w}_{O,i},\tilde{h}_{O,i}\right)\right| \\ \mathcal{L}_{\text{ang.}} &= \frac{1}{N^p}\sum_{i=1}^{N^p}(1-c_i)\left|\hat{\theta}_i-\tilde{\theta}_i\right|\end{aligned} \tag{18}$$

in which N^p is the total count of objects engaged in the pth epoch, $\hat{w}_{O,i}$, $\hat{h}_{O,i}$, and $\hat{\theta}_i$ come from the prediction of the model, and $\tilde{w}_{O,i}$, $\tilde{h}_{O,i}$, and $\tilde{\theta}_i$ are the pseudo-labels.

In terms of the tracking loss, our objective is to minimize the embedding distances between instances with the same identifier across adjacent frames, while maximizing the distances between instances with different trajectories. To achieve this, we employ an interframe associative embedding loss (AE loss) [49], which provides supervision for the embedding learning process

$$\begin{aligned}\mathcal{L}_{\text{pull}} &= \frac{1}{\min\left(N_{t-1}^p, N_t^p\right)}\sum_{i=1}^{\min\left(N_{t-1}^p,N_t^p\right)}\left(e_i^{t-1}-e_i^t\right)^2 \\ \mathcal{L}_{\text{push}} &= \frac{1}{N_{t-1}^p\cdot N_t^p}\sum_{j=1}^{N_{t-1}^p}\sum_{i=1,i\neq j}^{N_t^p}\max\left(0,\Delta-\left|e_j^{t-1}-e_i^t\right|\right) \\ \mathcal{L}_{\text{emb.}} &= \mathcal{L}_{\text{pull}}+\delta\mathcal{L}_{\text{push}}\end{aligned} \tag{19}$$

in which N_t^p is the number of objects in frame t in the pth epoch, e_i^t is the output embedding vector of the ith object in frame t, and δ is the weight coefficient.

V. Experimental Results and Analysis

In this section, extensive experiments were conducted on satellite video datasets from the Jilin-1 and Skysat satellites to validate the effectiveness of the proposed method.

A. Datasets

Specifically, we annotated a set of 15 videos from the Jilin-1 and Skysat satellite sources with OBBs and tracking IDs and employed them for training and evaluating the model performance.

1) Jilin-1 Dataset: The Jilin-1 satellite constellation, developed by Chang Guang Satellite Technology Company Ltd., includes optical video satellites capable of capturing high spatiotemporal resolution Earth observation video data. Nine satellite videos (001–009) captured by the Jilin-1 satellite are employed for quantitative evaluation, which covers three distinct areas: Dubai, UAE; San Diego, USA; and Frankfurt, Germany. These videos feature a size of about 1000×1000 pixels and a GSD of 0.92 m. Each video contains 100 frames with a frame rate of 10 frames/s, and they collectively depict 3448 different objects that appear and are labeled 1 17 982 times. Most objects fall within the size range of 8×8–12×12 pixels. Objects in the Dubai and San Diego were annotated with OBBs and tracking IDs based on the HBB labels from Feng et al. [46], while all objects in the Frankfurt videos were annotated anew using roLabelImg.[1] This dataset presents significant challenges, including small object size, dense distribution, complex motion patterns, and occlusions. We used videos 002, 006, and 009 for testing and the remaining for training.

TABLE I
Detailed Information of Satellite Videos

Platform	Area	ID	Size	Instances	Split
Jilin-1 (100 frames, 10FPS, 0.92m GSD)	Dubai	001	1000×1000	6,294	train
		002	1000×1000	16,646	test
		003	1000×1000	7,389	train
	San Diego	004	1500×700	7,860	train
		005	1000×1000	19,336	train
		006	1000×1000	17,158	test
	Frankfurt	007	1000×1000	12,935	train
		008	1000×1000	14,358	train
		009	1000×1000	16,006	test
Skysat (120 frames, 30FPS, 0.1m GSD)	Las Vegas	010	400×400	5,697	train
		011	400×600	4,761	test
		012	400×500	3,863	train
		013	400×600	2,906	train
		014	400×400	7,862	train
		015	600×500	5,546	test

2) Skysat Dataset: Skysat is a constellation of high-resolution imaging satellites operated by Planet Labs, capable of capturing both high-resolution still imagery and full-motion video of the Earth's surface. Owing to its video capabilities, Skysat has become a benchmark dataset for satellite video-based vehicle monitoring and motion analysis. Six videos from the Skysat satellite (010–015) were annotated, captured over Las Vegas, USA, on March 25, 2014. These videos have a size of approximately 400×600 pixels and a GSD of 1.0 m. Each video consists of 120 frames with a frame rate of 30 frames/s, totaling 30 635 instances from 828 distinct objects. Based on the HBB annotation by Zhang et al. [50], objects in videos 010–012 were annotated with OBBs and tracking IDs, while all objects in the remaining videos were annotated from scratch. This dataset poses several challenging factors, such as small objects, low contrast, frame misalignment, and frequent occlusions. In our experiments, we allocated videos 011 and 015 for testing and the rest for training.

Further details of these satellite videos are presented in Table I.

B. Implementation Details

We developed the P-OVDT in Python 3.9 using PyTorch 1.12.1, without employing pretrained parameters. The training

[1] roLabelImg: https://github.com/cgvict/roLabelImg

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TABLE II
DETECTION RESULTS OF DIFFERENT DETECTORS. THE BOLD VALUES REPRESENT THE BEST RESULTS FOR EACH ANNOTATION MANNER. ↓ INDICATES LOWER IS BETTER, ↑ INDICATES HIGHER IS BETTER

Method	Jilin-1				Skysat				FPS↑
	Recall↑	Precision↑	F1↑	AP↑	Recall↑	Precision↑	F1↑	AP↑	
Fully-supervised									
Rotated FCOS [52]	58.64	61.23	59.91	53.97	55.04	57.60	56.29	51.15	28.6
Rotated RetinaNet [53]	74.30	67.10	68.85	65.18	73.85	69.99	71.42	64.52	23.8
S^2A-Net [22]	63.05	67.45	65.25	60.24	73.00	68.74	70.81	64.46	21.2
Oriented-DETR [26]	59.60	65.09	62.20	58.82	58.25	61.97	60.11	57.51	8.3
Oriented R-CNN [2]	**76.26**	**71.14**	**73.94**	**70.14**	**77.61**	**73.27**	**75.38**	**67.57**	21.5
HBox-supervised									
BoxInst-RBox [54]	52.97	40.16	45.68	42.15	55.91	44.65	49.64	45.07	1.8
H2RBox [32]	56.52	51.62	53.96	44.03	58.29	49.64	53.62	47.35	22.4
H2RBox-v2 [33]	**71.73**	**63.92**	**67.60**	**55.93**	**69.18**	**60.55**	**64.58**	**53.67**	23.2
Point-supervised									
PointOBB [12]	46.89	41.86	43.87	41.96	45.79	41.52	43.16	40.29	23.0
PointOBB-v3 [55]	48.50	43.52	45.88	42.47	40.91	43.43	42.12	40.48	20.7
Point2RBox [11]	60.44	50.96	55.30	48.94	51.71	46.59	49.02	46.07	32.5
Point2RBox-v2 [56]	58.59	53.63	56.00	49.21	52.59	48.73	50.60	48.58	32.5
P-OVDT (Ours)	**72.30**	**65.54**	**68.76**	**59.68**	**67.90**	**60.48**	**63.98**	**55.79**	14.7

of P-OVDT was carried out on two RTX 3090 GPUs and an Intel i9-10940X CPU. The model was initiated with PyTorch's default configurations. Hourglass-104 [51] was selected as the backbone architecture, modifying the feature extraction process to operate on feature maps downsampled by 2×, as opposed to the standard 4× downsampling, to better preserve the features of small objects.

We employed the Adam optimizer to train P-OVDT for 35 epochs, with a batch size of 4. The initial learning rate was set to 0.001, and it was reduced by a factor of 10 at the 20th epoch. The initial difficulty threshold λ_0 is set to 0.2, and the coefficients in the loss function, δ, α, β, and γ, are, respectively, set to 0.1, 0.1, 0.5, and 1. The network was configured to process input images of 1024 × 1024 resolution, with random flipping serving as the sole data augmentation strategy during training. For data association, we employ ByteTrack [21] to get the object trajectories through the similarity of predicted ReID embeddings.

C. Main Results

1) Detection Results of Different Methods: We performed an extensive comparison of P-OVDT with various oriented object detectors. Given the nascent stage of exploration in single-point supervised oriented object detection methods, we also assessed fully supervised detectors and those based on HBBs for a more comprehensive analysis. The fully supervised detectors comprised Rotated RetinaNet [53], Rotated FCOS [52], Oriented R-CNN [2], S^2A-Net [22], and Oriented-DETR [26]. For HBB-based detection, we included BoxInst [54], H2RBox [32], and H2RBox-v2 [33], while PointOBB series [12], [55] and Point2RBox series [11], [56] represented point supervised detectors. The hyperparameters, such as feature map sampling rates and anchor sizes, were adjusted to better suit the satellite video scenarios.

The experimental results are detailed in Table II. P-OVDT emerged as the top performer under consistent annotation conditions, achieving $F1$-scores of 68.76% and 63.98% and APs of 59.68% and 55.79% on the Jilin-1 and Skysat datasets, respectively, significantly outperforming other point-supervised methods by approximately 10%–20% in terms of metrics. This performance gain can be largely attributed to our effective integration of temporal and visual cues for generating high-quality pseudo-labels and synthetic samples. Both the PointOBB and Point2RBox series face challenges in capturing the size and orientation of small, low-resolution objects. Compared to the HBB-based detector H2RBox-v2, P-OVDT also achieved near-par or marginally superior results. Among fully supervised methods, Oriented R-CNN delivered the highest AP values of 70.14% and 67.57%. P-OVDT notably reduced annotation costs while experiencing only a less than 20% performance degradation, which we consider a favorable tradeoff.

Fig. 9 illustrates the visual detection results in satellite videos, with true positives depicted in blue, false negatives in yellow, and false positives in red. PointOBB and Point2RBox show a significant number of false positives due to inaccurate size and angle predictions. Particularly in Fig. 9(c), most false detections exhibit similar angle deviations, attributed to feature drift that arises from the imprecise pseudo-labels. Conversely, P-OVDT, leveraging its ability to embed visual patterns in small vehicles and the curriculum learning strategy that progressively incorporates more reliable samples, demonstrates superior performance, consistent with its recognition as the top-performing point supervised detector according to the quantitative results in Table II.

2) Tracking Results of Different Methods: As our problem setting, P-OVDT is fundamentally an unsupervised learning approach for tracking tasks. Consequently, a comparison between P-OVDT and fully supervised trackers such as

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TABLE III

TRACKING RESULTS OF DIFFERENT TRACKERS. THE BOLD VALUES REPRESENT THE BEST RESULTS FOR EACH DATASET

Dataset	Method	MOTA↑	IDF1↑	IDP↑	IDR↑	MT↑	ML↓	FP↓	FN↓	IDs↓	FPS↑
Jilin-1	PointOBB-SORT	28.48	24.31	21.89	27.32	302	467	19614	15646	361	18.9
	PointOBB-Bytetrack	28.49	29.43	27.81	31.24	399	256	19571	15806	241	12.4
	Point2RBox-SORT	39.76	30.54	28.96	32.29	202	552	17440	12196	368	26.8
	Point2RBox-Bytetrack	39.40	34.48	31.53	38.04	326	370	17502	12427	252	18.5
	P-OVDT-SORT (Ours)	52.57	39.72	38.00	41.60	372	407	13510	9744	369	12.9
	P-OVDT-Bytetrack (Ours)	52.90	43.99	43.83	44.16	408	377	13596	9605	259	8.3
	P-OVDT-ReID-Bytetrack (Ours)	**53.84**	**55.38**	**50.85**	**60.78**	**499**	**141**	**13347**	**9508**	**137**	7.6
Skysat	PointOBB-SORT	27.18	22.63	21.09	24.42	89	98	4033	3374	97	19.4
	PointOBB-Bytetrack	29.50	30.03	29.69	30.38	91	80	3851	3352	61	14.0
	Point2RBox-SORT	33.29	26.33	25.94	26.74	60	119	3727	3059	88	27.5
	Point2RBox-Bytetrack	34.08	30.48	29.38	31.67	94	83	3760	2971	63	20.4
	P-OVDT-SORT (Ours)	46.32	42.21	39.60	37.29	73	97	3167	2291	73	13.6
	P-OVDT-Bytetrack (Ours)	47.03	49.43	45.58	42.28	117	58	3142	2260	57	9.8
	P-OVDT-ReID-Bytetrack (Ours)	**48.40**	**50.98**	**49.12**	**47.39**	**135**	**42**	**3025**	**2257**	**36**	9.1

TABLE IV

MODEL PERFORMANCE WITH DIFFERENT PSEUDO-LABEL GENERATION METHODS

Generation Methods	Jilin-1		Skysat	
	AP↑	MOTA↑	AP↑	MOTA↑
Baseline w/ Blob Segmentation	8.59	-7.65	6.48	-4.54
Baseline w/ Sequence-Aligned	20.83	14.07	17.32	12.77
P-OVDT w/ Blob Segmentation	23.40	17.32	24.13	18.90
P-OVDT w/ Sequence-Aligned	**59.68**	**53.84**	**55.79**	**48.40**

DeepSORT [57], FairMOT [17], and CenterTrack [18] would be inequitable. Meanwhile, these methods do not support OBBs. To maintain fairness, we integrated the detection results of point-supervised detectors (PointOBB and Point2RBox) with classic unsupervised trackers like SORT [20], and ByteTrack [21] to obtain object association outcomes, thereby validating the efficacy of the proposed method.

As Table III illustrates, when employing ByteTrack as the tracker and motion information as the similarity measure, P-OVDT achieves a significant improvement over Point2RBox. Specifically, P-OVDT's MOTA score increases by approximately 13% on both datasets, with corresponding IDF1 improvements of 9.51% and 20.37%, respectively. P-OVDT also demonstrated significant advantages in metrics such as MT, ML, and IDs, and the gap in performance compared to PointOBB was more pronounced. Moreover, when integrating predicted ReID embeddings for the similarity measure, the performance was further enhanced, achieving MOTA of 53.84% and 48.40% and IDF1 of 55.38% and 50.98% on the two datasets. These gains are achieved at an inference speed of 7.6 and 9.1 frames/s on the two datasets, respectively—despite the additional overhead of jointly learning ReID embeddings. On the Jilin-1 dataset, IDF1 increased by 11.39%, further underscoring the effectiveness of our end-to-end framework in learning discriminative appearance and motion.

Fig. 10 visualizes tracking trajectories, with more complete and smooth trajectories indicating more accurate predictions. The high-quality trajectories produced by P-OVDT can be attributed to the proposed spatiotemporal consistency metric and curriculum learning strategy, which together suppress noisy samples and promote stable trajectory formation. It is evident that the limitations of the detectors result in fragmented trajectories, particularly pronounced in the case of PointOBB. These fragment trajectories further elevate the likelihood of ID switches. In contrast, P-OVDT generates relatively complete trajectories, and the position offsets of instances are notably smaller than other methods. While these results are not flawless, considering our minimal annotation cost, we find them to be satisfactory.

D. Ablation Study and Analysis

The effectiveness of the various modules within P-OVDT was rigorously assessed through ablation experiments. Various combinations of these modules were integrated into the baseline to ascertain their contribution. Tables IV and V present the outcomes of these experiments.

1) Sequence-Aligned Pseudo-Label Generation: To evaluate the effectiveness of our proposed pseudo-label generation method, we compared our results with those obtained from training using pseudo-labels generated by foreground blob segmentation (as shown in Fig. 4), and the experimental results are recorded in Table IV. In the first two rows, we compared the two methods based on the baseline model. It is evident that the method utilizing blob segmentation hardly provides accurate pseudo-labels for the network to perform weakly supervised learning, achieving less than 10% AP and a negative MOTA. In the last two rows, we used the complete P-OVDT pipeline, merely replacing the pseudo-label generation method for comparison. In this case, the method based on blob segmentation experienced a performance decline of more than half, proving that these inaccurate noise samples severely interfere with feature learning.

2) Visual-Aligned Synthetic Objects: To evaluate the effectiveness of synthetic objects, we conducted experiments under different sample conditions, with the results presented in the first two rows of Table V. It can be seen that using pseudo-labels alone yielded APs of only 20.83% and 17.32% on the two datasets. While combining the synthetic objects markedly enhanced APs by 24.52% and 36.19%, highlighting the complementary nature of the samples in providing diverse

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TABLE V
PERFORMANCE OF THE ABLATION EXPERIMENT IN P-OVDT

Pseudo Labels	Synthetic Objects	Curriculum Learning	Spatial Pooling Module	Jilin-1					Skysat				
				AP↑	MOTA↑	IDF1↑	MT↑	ML↓	AP↑	MOTA↑	IDF1↑	MT↑	ML↓
✓				20.83	14.07	7.31	163	652	17.32	12.77	6.62	56	143
✓	✓			45.35	37.24	36.13	358	330	43.51	37.30	28.75	49	135
✓	✓	✓		59.44	51.40	52.75	480	190	53.13	44.47	47.15	110	45
✓	✓	✓	✓	**59.68**	**53.84**	**55.38**	**499**	**141**	**55.79**	**48.40**	**50.98**	**135**	**42**

object features and more accurate label information, thus significantly boosting weakly supervised learning outcomes.

Fig. 11 depicts the visualization results of the data generation module for Video-009 and Video-012, with noise samples highlighted by yellow dashed boxes. It is clear that, despite careful pseudo-label generation, Fig. 11(a) shows instances where the rotation angle is imprecise. In Fig. 11(b), it is apparent that accurately estimating the sizes of OBBs is challenging. In contrast, synthetic objects possess precise pseudo-labels. It is noteworthy that synthetic objects also exhibit errors, predominantly due to inaccurate background extraction. Consequently, P-OVDT leverages the complementary nature of pseudo-labeled real objects and accurately labeled synthetic objects to guide the feature learning process.

3) Curriculum Learning: By the comparison between the second and third rows in Table V, the curriculum learning schedule has led to a 14.09% and 10.59% increase in APs and 16.16% and 8.56% improvement in MOTAs. This advancement is attributed to the spatiotemporal consistency measure, which effectively gauges the complexity of the samples. By directing the learning process from simpler to more complex aspects, the measure minimizes the interference caused by numerous noisy samples during the early stages of training, thereby enhancing the overall learning efficiency.

Moreover, it is worth noting that sample generation and learning methods are primarily employed during preprocessing or the training phase, and therefore do not contribute to inference latency.

4) Spatial Pooling Module: In the embedding branch, P-OVDT employs a newly developed SPM to extract distribution patterns between objects. When comparing the final two rows of Table V, it becomes evident that this approach significantly enhances the MOTA by 2.44% and 2.54% and increases IDF1 by 2.63% and 3.83% on both datasets, respectively. Despite these gains, the SPM adds only a modest 3.8% (7.2M) to the backbone's parameter count and negligible inference time overhead (less than 1 frames/s). This underscores the effectiveness of the spatial distribution pattern in assisting the prediction of discriminative embeddings, thereby mitigating the impact of the absence of appearance features in vehicle objects.

E. Sensitivity Analysis

In this section, we assess the sensitivity of P-OVDT by adjusting key parameters and modules within the framework.

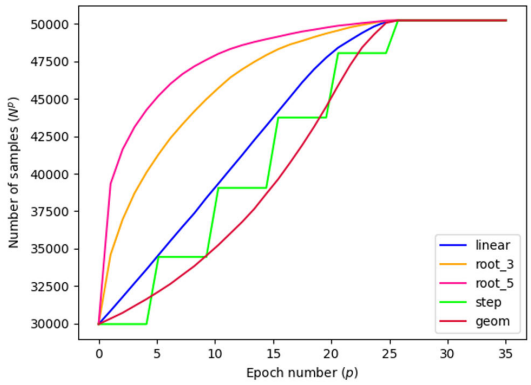


Fig. 12. Trend in the number of training samples evolving. It is evident that the *geometric progression function* allows simple samples more training time, while the root function does the opposite. The *step function*, which abruptly incorporates a large number of samples, can introduce instability to the convergence process.

1) Curriculum Learning Pacing Functions: The pacing function $\lambda(p)$ maps the training epoch number p to a scalar λ within the range (0, 1], thereby determining the samples engaged in training at each epoch. Alongside the *geometric progression function*, we experimented with various classic scheduler functions, such as *step functions*, *linear functions* [58], and *root functions* [59], wherein the *linear functions* are defined as follows:

$$\lambda_{\text{linear}}(p) = \min\left(1, \lambda_0 + \frac{1-\lambda_0}{p_{\max}} \cdot p\right). \tag{20}$$

And the *root functions* are defined as follows:

$$\lambda_{\text{root-}r}(p) = \min\left(1, \sqrt{\frac{1-\lambda_0^r}{p_{\max}} \cdot p + \lambda_0^r}\right) \tag{21}$$

where r is used to control the sharpness of the curve. It is worth noting that, unlike traditional curriculum learning methods, we do not interpret λ as the proportion value of samples engaged in training but utilize it as a threshold for the spatial–temporal consistency measure. Fig. 12 illustrates the evolution of the number of samples engaged in training as the process progresses on the Skysat dataset, with the Jilin-1 dataset exhibiting a similar trend.

The experimental outcomes, summarized in Table VI and illustrated by the sample progression curves in Fig. 12, reveal distinct behaviors across pacing functions. The linear function shows a steady increase in sample count, supporting stable feature learning. In contrast, the step function introduces

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TABLE VI
MODEL PERFORMANCE WITH DIFFERENT PACING FUNCTIONS

Pacing Functions	Jilin-1		Skysat	
	AP↑	MOTA↑	AP↑	MOTA↑
Step	53.65	52.79	50.95	46.23
Liner (Eq. (20))	58.81	53.05	54.53	**48.42**
Root-3 (Eq. (21))	57.22	52.97	53.78	47.89
Root-5 (Eq. (21))	52.07	51.53	49.85	47.40
Geometric (Eq. (16))	**59.68**	**53.84**	**55.79**	48.40

TABLE VII
MODEL PERFORMANCE WITH DIFFERENT λ_0

λ_0	Jilin-1		Skysat	
	AP↑	MOTA↑	AP↑	MOTA↑
0	59.28	53.03	55.32	47.89
0.2	**59.68**	**53.84**	**55.79**	**48.40**
0.4	58.62	52.70	55.10	48.43
0.6	57.61	53.43	53.81	47.96

abrupt sample increments at fixed intervals, causing training instability. The root functions promote rapid early expansion but introduce noisy samples prematurely, hindering initial feature extraction. Finally, the geometric progression begins with simpler samples and gradually incorporates more complex ones, aligning with an easy-to-hard training strategy. Among these, the geometric progression achieves the highest AP and MOTA, as validated in Table VI, demonstrating its effectiveness in maintaining feature stability and improving overall performance. Consequently, the performance of using *step functions* is even worse than learning without a curriculum learning approach (refer to Table V).

2) Initial Difficulty Threshold λ_0: In P-OVDT, the effect of curriculum learning is also influenced by the initial difficulty threshold λ_0. In an ideal scenario, synthetic objects possess perfectly accurate labels; thus, the spatiotemporal consistency measure for these objects is set as a manually defined hyperparameter λ_0, which ensures that they are always included in the training process from the outset. That is to say, λ_0 governs the percentage of synthetic objects included in the initial phase of training. As shown in Table VII, when λ_0 is 0, it implies that only synthetic objects are involved in training during the initial epochs, with no real objects; conversely, as λ_0 increases, the proportion of real objects incorporated into the training during the initial epochs also progressively grows.

VI. DISCUSSION

Despite the promising results achieved by P-OVDT, we acknowledge several limitations of our current work, which also point to valuable directions for future research.

The performance of the proposed framework may be limited in highly dense or occluded scenes, where unreliable foreground extraction and disrupted motion-based association can lead to fragmented tracklets and noisy pseudo-labels. Furthermore, while designed for satellite videos, its generalization to different domains, such as UAV or street-level videos, remains unverified, and domain shifts in object appearance and motion patterns may impair pseudo-label generation and feature encoding. Additionally, the accuracy of pseudo-OBB estimation is sensitive to motion alignment and GSD parameters; any inaccuracies in these can degrade orientation and size estimates, further reducing pseudo-label quality.

Addressing these limitations opens up new research directions. We plan to integrate more robust association models to complement motion cues in handling occlusions. Investigating domain adaptation techniques or designing a more viewpoint-invariant architecture could enhance the model's applicability across different platforms. Furthermore, exploring self-supervised methods to estimate motion parameters and GSD directly from the video data could make the framework more autonomous.

VII. CONCLUSION

To maximize the annotation cost of vehicle dynamic analysis, we present a framework to detect and track oriented vehicles in satellite videos using identifier-free point annotations. The proposed end-to-end network, P-OVDT, significantly enhances model performance in both sample acquisition and the learning process. P-OVDT innovatively generates high-quality pseudo-labels by exploiting temporal information across video sequences. Additionally, it introduces a customized Gaussian distribution-based strategy to simulate vehicle appearance patterns, aiding in weakly supervised learning. During training, P-OVDT assigns lower difficulty scores to objects with consistent labels in long sequences through a novel spatiotemporal consistency criterion, allowing the network to learn from easy to more challenging examples. Extensive experimental results on two satellite video datasets validate the superiority of our proposed method. We aspire that this work paves the way for new research directions in weakly supervised satellite video object detection and tracking.

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